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CardFlow Credit Origination Platform

Multi-Tenant SaaS Solution for Credit Card Application and Approval Automation

Functional Design Document

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Version History

Rev #	System Version	Description	Author	Date	Reviewer	Date
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1. Introduction

This section introduces the purpose, scope, and references of the Functional Design Document, providing context for the functional design of the **CardFlow Credit Origination Platform**.

1.1. Purpose of the Document

The purpose of this Functional Design Document (FDD) is to define how the **CardFlow Credit Origination Platform** operates from a functional perspective. This document translates the business requirements defined in the Business Requirements Document (BRD) and the system behaviors specified in the Software Requirements Specification (SRS) into detailed functional workflows, decision logic, and process flows.

It provides a structured representation of how the system processes credit applications, manages workflow transitions, and enforces business rules across different stages of the application lifecycle.

1.2. Scope of Functional Design

This document covers the functional design of the core workflows and system behaviors in the **CardFlow Credit Origination Platform**. It includes the processing of credit applications, amendment requests, and reinstatement requests, as well as the associated validation, review, routing, and approval mechanisms.

The document focuses on how the system behaves in response to user actions and defined business rules. Technical implementation details such as database design, APIs, and infrastructure are outside the scope of this document and are addressed in the Technical Design Document (TDD).

1.3. Intended Audience

This document is intended for stakeholders involved in the design, development, testing, and validation of the system, including:

- Business Analysts
- System Designers
- Developers
- Test Engineers
- Project Stakeholders

It serves as a reference for understanding functional workflows and system behavior during implementation and quality assurance activities.

1.4. Definitions and Terminology

This document uses terminology consistent with the definitions established in the Business Requirements Document and Software Requirements Specification. Key terms related to credit application processing, workflow stages, and user roles are defined in the respective documents to ensure consistency across all documentation artifacts.

1.5. References

The following documents are referenced in this Functional Design Document (FDD):

- Business Requirements Document (BRD)
- Software Requirements Specification (SRS)

These documents provide the business context and system requirements that form the basis of the functional design described in this document.

2. Functional Overview

This section provides a high-level overview of the system's functional behavior and key capabilities in the **CardFlow Credit Origination Platform**.

2.1. System Functional Summary

This subsection summarizes the core functions supported by the system and how they collectively enable the end-to-end credit application lifecycle.

The **CardFlow Credit Origination Platform** supports the end-to-end lifecycle of credit application processing in a multi-tenant SaaS environment. The system enables applicants and internal users to initiate, submit, review, and process credit-related requests through structured workflows.

Core system functions include:

- Credit application submission and processing
- Review and approval workflows
- Amendment of existing applications or account information
- Reinstatement of suspended or inactive credit accounts
- Document management and validation
- Role-based access and administrative configuration
- Reporting and audit tracking

These capabilities are supported through configurable workflows and business rules that allow tenant organizations to define their own approval structures and operational policies.

2.2. Key Functional Capabilities

This subsection outlines the major functional capabilities of the system, grouped by operational areas.

- **Application Processing**
    - Supports creation, submission, and lifecycle management of credit applications
    - Ensures validation of required data and supporting documents
- **Workflow Management**
    - Routes applications through configurable review and approval processes

    - Supports multi-level approval hierarchies

- **Amendment and Reinstatement**
    - Enables controlled modification of existing records
    - Supports reinstatement of inactive or suspended accounts
- **Document Handling**
    - Manages upload, storage, and retrieval of supporting documents
    - Ensures document validation and accessibility
- **Administration and Configuration**
    - Allows administrators to configure roles, permissions, and workflows
    - Supports dynamic adjustment of business rules
- **Reporting and Audit**
    - Provides visibility into application status and processing history
    - Maintains audit logs for traceability and compliance

2.3. Functional Architecture Overview

This subsection describes the high-level functional components of the system and how they interact to support workflow execution.

The system operates through the interaction of the following functional components:

- **User Interaction Layer** – Interfaces used by applicants, reviewers, approvers, and administrators
- **Workflow Engine** – Manages routing, state transitions, and approval logic
- **Validation Engine** – Enforces data completeness, format rules, and eligibility criteria
- **Document Management Component** – Handles storage and retrieval of supporting documents
- **Audit and Logging Component** – Captures system and user actions for traceability

These components work together to ensure consistent, controlled, and traceable processing of credit-related workflows across the platform.

3. Workflow Lifecycle Model

This section defines the standard workflow lifecycle used in the **CardFlow Credit Origination Platform**. It describes the sequence of stages that govern how credit applications and related requests are processed from initiation to outcome.

The lifecycle model establishes a consistent structure for all workflows in the system, ensuring that processing stages, decision points, and system behaviors are applied uniformly across application, amendment, and reinstatement processes.

3.1. Standard Workflow Lifecycle

This subsection defines the standard stages that make up the workflow lifecycle. Each stage represents a distinct phase in the processing of credit-related requests and is associated with specific system behaviors and user interactions.

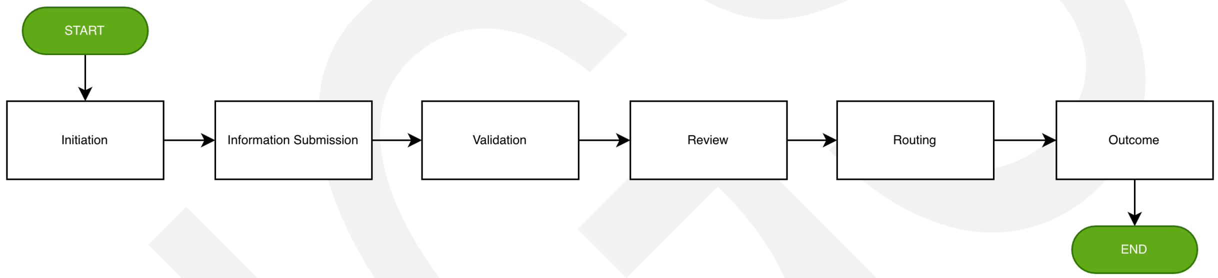

Figure 1 CardFlow Credit Origination Platform Workflow Lifecycle

Table 1 CardFlow Credit Origination Platform Workflow Lifecycle

Workflow Lifecycle	Definition
Initiation	The process begins when an authorized user initiates a new request, such as a credit application, amendment, or reinstatement.
Information Submission	Required information and supporting documents are provided by the user to support the request.
Validation	The system evaluates submitted data to ensure completeness, correctness, and compliance with defined validation rules.

Workflow Lifecycle	Definition
Review	Authorized reviewers assess the submitted information, supporting documents, and eligibility conditions.
Routing	The system determines and assigns the appropriate reviewers or approvers based on configured workflow rules and approval hierarchies.
Outcome	The request reaches a final or intermediate decision, such as approval, rejection, or return for additional information.

3.2. Workflow States Definition

This subsection defines the system states associated with each stage of the workflow lifecycle. These states represent the status of a request as it progresses through the system.

Table 2 Workflow States Definition

Workflow State	Definition
Draft	The request is initiated but not yet submitted.
Submitted	The request has been submitted and is awaiting validation.
Under Validation	The system is validating submitted information and documents.
Under Review	The request is being evaluated by reviewers.
Pending Approval	The request is awaiting final decision from an authorized approver.
Approved	The request has been approved.
Rejected	The request has been rejected and is no longer processed further.

Workflow State	Definition
Returned	The request is returned to the initiating user for correction or additional information.

3.3. State Transition Rules

This subsection defines how requests transition between workflow states based on system actions, user actions, and decision outcomes.

- **Draft → Submitted**
- **Submitted → Under Validation**
- **Under Validation → Under Review**
- **Under Review → Pending Approval**
- **Pending Approval → Approved / Rejected / Returned**
- **Returned → Submitted**

Key Transition Logic

- A request moves from **Draft** to **Submitted** when the user completes and submits required information.
- A request moves to **Under Validation** once submitted.
- A request proceeds to **Under Review** after successful validation.
- A request enters **Pending Approval** after review is completed.
- A request reaches **Approved**, **Rejected**, or **Returned based** on decision outcomes.
- A **Returned** request may be corrected and resubmitted.

Note:

Returned applications re-enter the workflow through resubmission and are processed again through validation, review, and approval as applicable.

Table 3 State Transition Rules

Workflow Cycle	State	Condition/Trigger	Next State
Initiation	**Draft**	User submits request	**Submitted**
Information Submission	**Submitted**	System receives submission	**Under Validation**
Validation	**Under Validation**	Validation successful	**Under Review**
Validation	**Under Validation**	Validation failed	**Returned**
Review	**Under Review**	Review completed	**Pending Approval**
Review	**Under Review**	Reviewer requests changes	**Returned**
Routing	**Pending Approval**	Approver approves	**Approved**
Routing	**Pending Approval**	Approver rejects	**Rejected**
Routing	**Pending Approval**	Approver returns for correction	**Returned**
Outcome	**Returned**	User updates and resubmits	**Submitted**

3.4. Status Mapping Across Workflows

The standard workflow states are consistently applied across:

- Credit Application Workflow
- Amendment Workflow
- Reinstatement Workflow

While specific processing steps may vary, all workflows follow the same lifecycle structure and utilize the same set of system states to ensure consistency in tracking, reporting, and audit.

4. Core Functional Workflows

This section defines the detailed functional behavior of the core workflows supported by the **CardFlow Credit Origination Platform**.

4.1. Credit Application Workflow

This subsection describes the functional design of the Credit Application Workflow, including the sequence of activities, decision points, state transitions, and exception handling that govern how a credit application is processed in the system.

4.1.1. Workflow Overview

The Credit Application Workflow represents the primary operational process in the platform. It begins when an authorized user initiates a new application and ends when the application reaches a final decision or is returned for correction.

This workflow follows the standard **Figure 1 CardFlow Credit Origination Platform Workflow Lifecycle**.

The workflow supports controlled processing of application data, document validation, review activities, approval routing, and outcome determination in accordance with defined business rules and approval hierarchies.

4.1.2. Swimlane Diagram

This subsection presents the workflow from a role-based perspective, showing how activities are distributed across the tenant user, applicant, the system, reviewers, and approvers.

The following Swimlane diagram expands the high-level workflow defined in the Business Requirements Document by incorporating system-level validation, decision logic, and exception handling.

Note:

- The diagram provides a high-level functional representation of how responsibilities are distributed across participating roles.
- Detailed transition logic and exception behavior are defined in the subsections that follow.

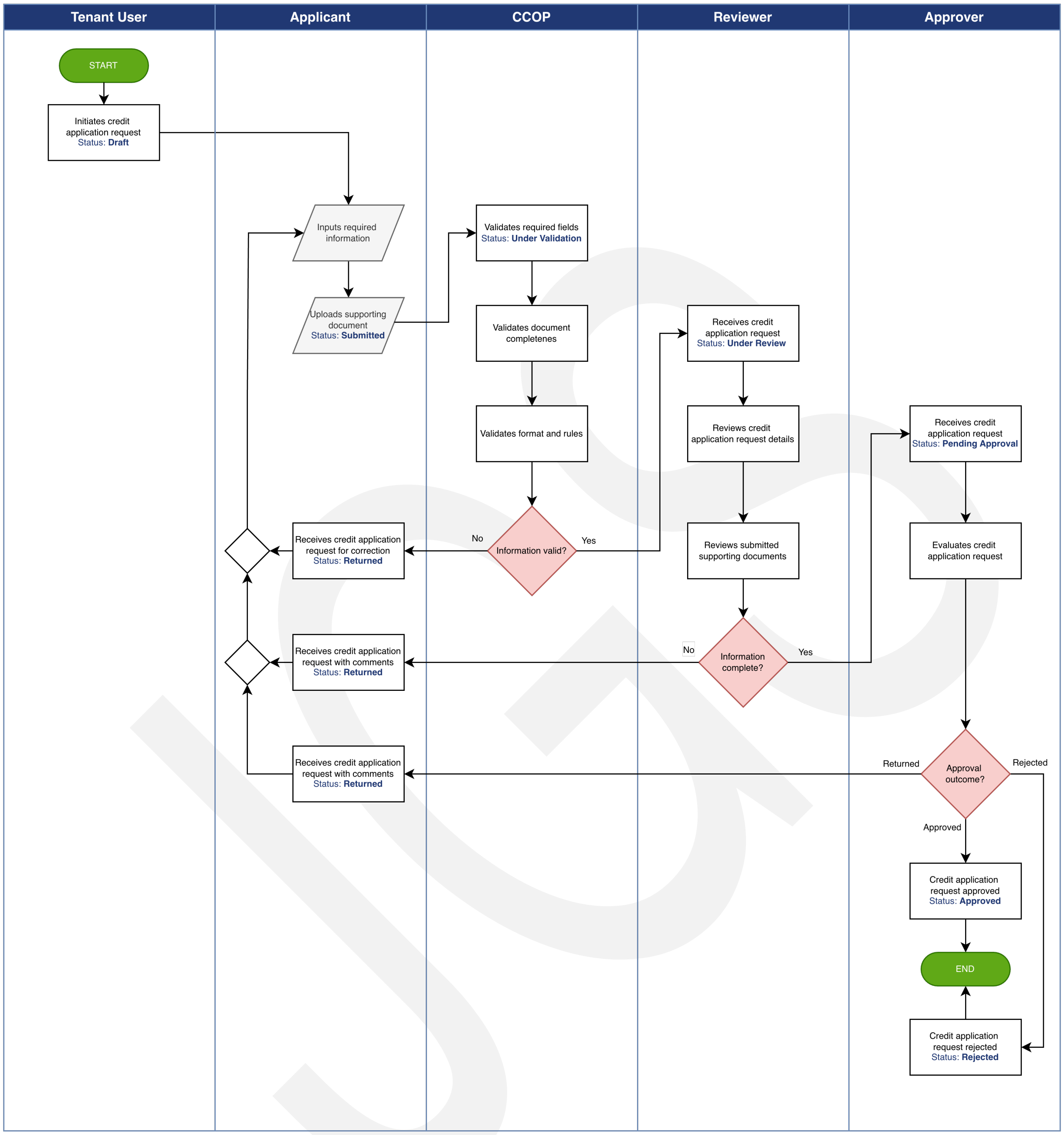

Figure 2 Credit Application Workflow

4.1.3. Step-by-Step Functional Behavior

This subsection describes how the system behaves at each stage of the Credit Application Workflow.

Table 4 Credit Application Workflow Functional Behavior

Step	Workflow Lifecyle	Functional Behavior
1	Application Initiation	An authorized tenant user initiates a new credit application. At this stage, the request is created in **Draft** state.
2	Information Submission	• The applicant provides required application details and uploads supporting documents. • The application remains editable until the user submits it for processing.
3	Validation	• Upon submission, the system validates the completeness of required fields, verifies document presence, and checks applicable submission rules. • If validation fails, the application moves to **Returned** state for correction.
4	Review	• If validation succeeds, the application proceeds to reviewer evaluation. • Reviewers assess application data and supporting documents and may either continue the process or return the application for additional information.
5	Routing	When review is completed successfully, the system routes the application to the

Step	Workflow Lifecycle	Functional Behavior
		appropriate approver based on the configured approval workflow.
6	Outcome	• The approver determines the outcome: ○ **Approved** – The credit application request is approved. ○ **Rejected** – The credit application request is rejected. ○ **Returned** – Request is sent back for correction • The system updates the application status accordingly and records all relevant workflow actions for traceability.

4.1.4. Decision Points and Validation Logic

This subsection defines the key decision points that affect progression of the credit application workflow.

Table 5 Credit Application Workflow Decision Points and Validation Logic

Decision Point	Definition
Submission Validity	The system evaluates whether the submitted application includes all required information and supporting documents: • If valid, the application proceeds to review. • If invalid, the application is returned for correction.

Decision Point	Definition
Review Completion	The reviewer evaluates whether the submitted application is complete and suitable for progression: • If complete, the application proceeds to approval routing. • If incomplete, the application is returned to the initiating user.
Approval Outcome	The approver determines the result of the application. • If approved, the application moves to **Approved** state. • If rejected, the application moves to **Rejected** state. • If returned, the application moves to **Returned** state.

Validation Logic Summary

The following validation rules are applied before allowing the request to progress to the next workflow stage:

- Mandatory field completion
- Required document submission
- Supported file format and size checks
- Eligibility for further processing based on configured rules

4.1.5. State Transitions

This subsection describes how the Credit Application Workflow transitions through system states.

> **Note:**
>
> Returned applications re-enter the workflow through resubmission and are processed again through validation, review, and approval as applicable.

Table 6 Credit Application Workflow State Transitions

Workflow Cycle	Current State	Trigger / Condition	Next State
Initiation	**Draft**	Tenant User initiates credit application	**Draft**
Initiation	**Draft**	User submits application	**Submitted**
Information Submission	**Submitted**	System receives submission	**Under Validation**
Validation	**Under Validation**	Validation successful	**Under Review**
Validation	**Under Validation**	Validation failed	**Returned**
Review	**Under Review**	Review completed	**Pending Approval**
Review	**Under Review**	Reviewer requests changes	**Returned**
Routing	**Pending Approval**	Approver approves	**Approved**
Routing	**Pending Approval**	Approver rejects	**Rejected**
Routing	**Pending Approval**	Approver returns for correction	**Returned**
Outcome	**Returned**	User updates and resubmits	**Submitted**

4.1.6. Exception Scenarios

This subsection identifies common exception scenarios that may occur in the Credit Application Workflow and describes the expected functional response.

- **Incomplete Submission** – If mandatory fields or required documents are missing, the system prevents successful progression and returns the application for correction.

- **Invalid Document Upload** – If uploaded files do not meet format or size requirements, the system rejects the submission and displays an appropriate validation message.
- **Reviewer Return** – If the reviewer determines that additional information is required, the application is returned to the initiating user with review comments.
- **Approver Return** – If the approver requires clarification or correction, the application is returned to the initiating user and may be resubmitted after updates.
- **Rejected Application** – If the approver rejects the application, no further workflow progression occurs unless a new application is initiated.

4.2. Credit Amendment Workflow

This subsection describes the functional design of the Credit Amendment Workflow, including the sequence of activities, validation logic, decision points, and state transitions involved in processing amendment requests in the system.

4.2.1. Workflow Overview

The Credit Amendment Workflow enables authorized users to request changes to existing credit applications or account information. The workflow ensures that all amendments are validated, reviewed, and approved before changes are applied.

This workflow follows the standard **Figure 1 CardFlow Credit Origination Platform Workflow Lifecycle**.

> **Note:**
>
> Compared to the credit application workflow, the amendment process includes an **eligibility validation step** prior to submission to ensure that only valid records can be modified.

4.2.2. Swimlane Diagram

This subsection presents the workflow from a role-based perspective, showing how amendment processing responsibilities are distributed across the tenant user, applicant, the system, reviewers, and approvers.

The following Swimlane diagram expands the high-level workflow defined in the Business Requirements Document by incorporating system-level validation, decision logic, and exception handling.

Note:

- The diagram provides a high-level functional representation of how responsibilities are distributed across participating roles.
- Detailed transition logic and exception behavior are defined in the subsections that follow.

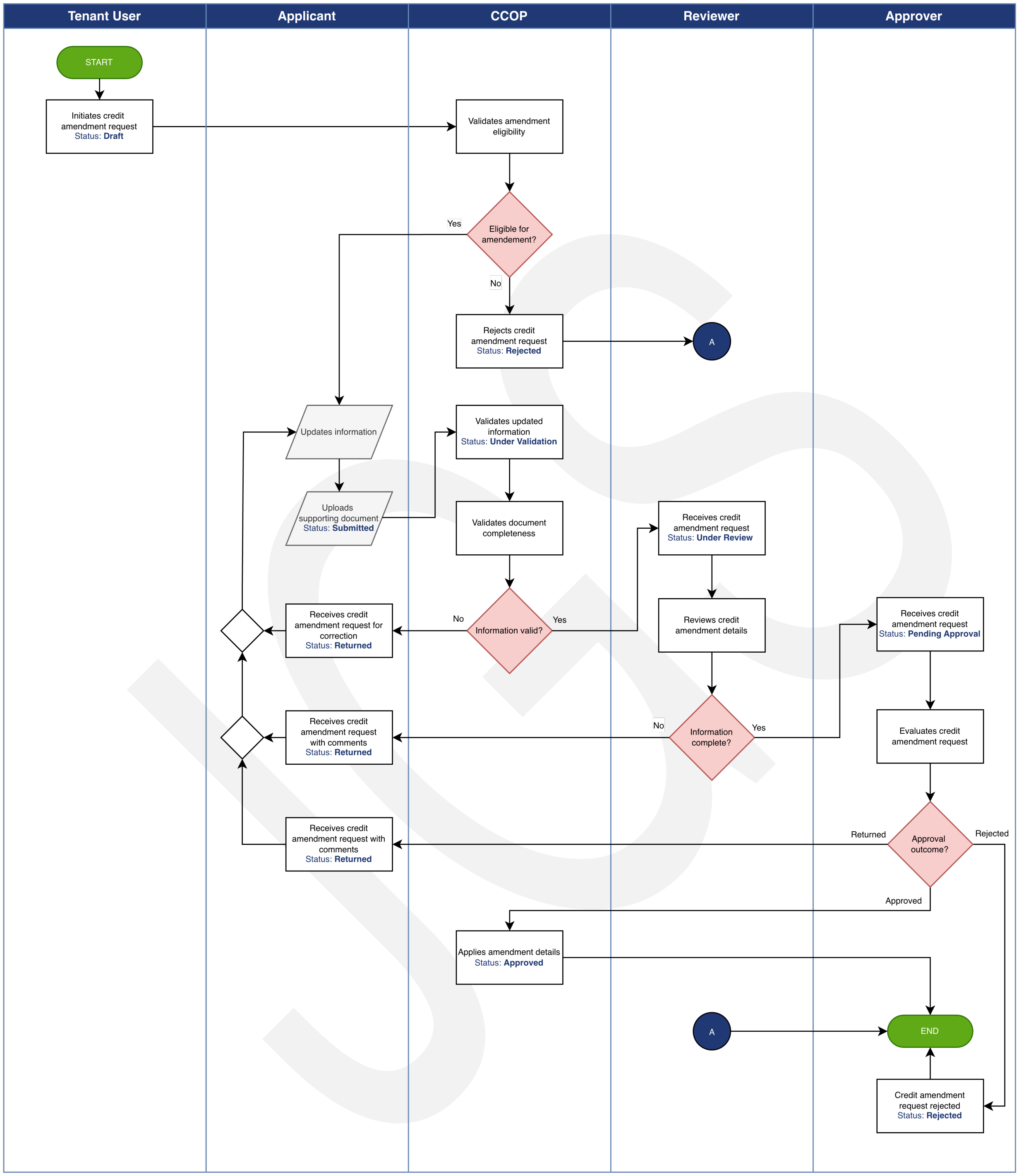

Figure 3 Credit Amendment Workflow

4.2.3. Step-by-Step Functional Behavior

This subsection describes the system behavior at each stage of the amendment workflow.

Table 7 Credit Amendment Workflow Functional Behavior

Step	Workflow Lifecyle	Functional Behavior
1	Amendment Initiation	An authorized tenant user initiates an amendment request for an existing credit application or account.
2	Eligibility Validation	• The system verifies whether the selected application or account is eligible for amendment based on defined business rules. • If the request is not eligible, it is rejected.
3	Information Submission	The applicant updates the required fields and uploads supporting documents relevant to the amendment.
4	Validation	• The system validates the completeness and correctness of updated information and supporting documents. • Invalid submissions are returned for correction.
5	Review	The reviewer evaluates the amendment request to ensure that all required updates are properly documented and justified.
6	Routing	The system routes the request to the appropriate approver based on configured approval workflows.

Step	Workflow Lifecyle	Functional Behavior
7	Outcome	• The approver determines the outcome: ○ **Approved** – Amendment is applied ○ **Rejected** – No changes are applied ○ **Returned** – Request is sent back for correction • The system records all relevant workflow actions for traceability.

4.2.4. Decision Points and Validation Logic

This subsection defines the key decision points that affect progression of the credit amendment workflow.

Table 8 Credit Amendment Workflow Decision Points and Validation Logic

Decision Point	Definition
Amendment Eligibility	Determines whether the selected record can be modified: • If eligible, the amendment requests proceeds to review. • If not eligible, the amendment request is returned for correction.
Submission Validity	Checks completeness and correctness of updated data: • If valid, the amendment requests proceeds to review. • If invalid, the amendment request is returned for correction.

Decision Point	Definition
Review Completion	Determines if the amendment request is sufficiently complete: • If complete, the amendment requests proceeds to approval routing. • If incomplete, the amendment request is returned for correction.
Approval Outcome	Decision by approver: • If approved, amendment is applied and requests moves to **Approved** state. • If rejected, the request is terminated and request moves to **Rejected** state. • If returned, the request is sent back for correction and moves to **Returned** state.

Validation Logic Summary

The following validation rules are applied before allowing the request to progress to the next workflow stage:

- Mandatory field completion for updated information
- Required document submission to support requested changes
- Supported file format and size checks for uploaded documents
- Eligibility validation to ensure the record is allowed for amendment
- Consistency checks between existing data and updated values
- Compliance with defined business rules for modification

4.2.5. State Transitions

This subsection describes how the Credit Amendment Workflow transitions through system states.

Note:

Returned applications re-enter the workflow through resubmission and are processed again through validation, review, and approval as applicable.

Table 9 Credit Amendment Workflow State Transitions

Workflow Cycle	Current State	Trigger / Condition	Next State
Initiation	**Draft**	Tenant User initiates amendment	**Draft**
Validation	**Draft**	Eligibility fails	**Rejected**
Information Submission	**Draft**	User submits updates	**Submitted**
Validation	**Submitted**	Validation successful	**Under Review**
Validation	**Submitted**	Validation failed	**Returned**
Review	**Under Review**	Review completed	**Pending Approval**
Review	**Under Review**	Reviewer returns	**Returned**
Routing	**Pending Approval**	Approver approves	**Approved**
Routing	**Pending Approval**	Approver rejects	**Rejected**
Routing	**Pending Approval**	Approver returns	**Returned**
Outcome	**Returned**	User resubmits	**Submitted**

4.2.6. Exception Scenarios

This subsection identifies common exception scenarios that may occur in the Credit Amendment Workflow and describes the expected functional response:

- **Ineligible Amendment** – If the selected record is not eligible for modification, the system rejects the request and terminates the workflow.
- **Invalid Submission** – If required updates or documents are missing or incorrect, the system returns the request for correction.

- **Reviewer Return** – If the reviewer identifies incomplete or insufficient information, the request is returned to the applicant.
- **Approver Return** – If additional clarification is needed, the request is returned for further updates.
- **Rejected Amendment** – If rejected, no changes are applied and the workflow ends.

4.3. Credit Reinstatement Workflow

This subsection describes the functional design of the Credit Reinstatement Workflow, including the sequence of activities, validation logic, decision points, and state transitions involved in processing reinstatement requests in the system.

4.3.1. Workflow Overview

The Credit Reinstatement Workflow enables authorized users to request reactivation of previously suspended or inactive credit accounts. The workflow ensures that reinstatement requests are validated, reviewed, and approved before the account is reactivated.

This workflow follows the standard **Figure 1 CardFlow Credit Origination Platform Workflow Lifecycle**.

> **Note:**
>
> Compared to the amendment workflow, reinstatement focuses on **account eligibility** and **approval**, **with fewer modification** steps.

4.3.2. Swimlane Diagram

This subsection presents the workflow from a role-based perspective, showing how reinstatement processing responsibilities are distributed across the tenant user, applicant, the system, reviewers, and approvers.

The following Swimlane diagram expands the high-level workflow defined in the Business Requirements Document by incorporating system-level validation, decision logic, and exception handling.

Note:

- The diagram provides a high-level functional representation of how responsibilities are distributed across participating roles.
- Detailed transition logic and exception behavior are defined in the subsections that follow.

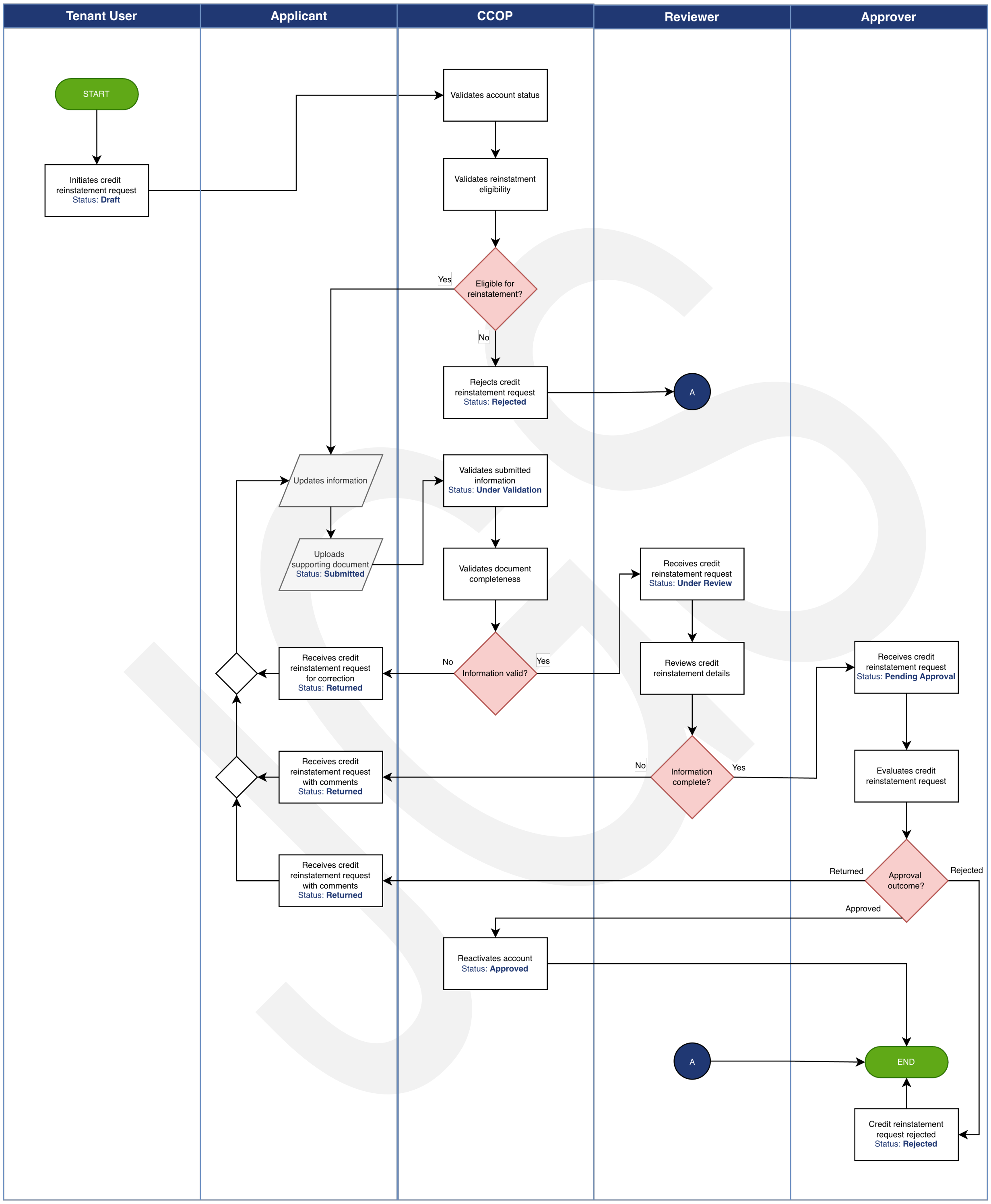

Figure 4 Credit Reinstatement Workflow

4.3.3. Step-by-Step Functional Behavior

This subsection describes the system behavior at each stage of the reinstatement workflow.

Table 10 Credit Reinstatement Workflow Functional Behavior

Step	Workflow Lifecyle	Functional Behavior
1	Reinstatement Initiation	An authorized tenant user initiates a reinstatement request for a suspended or inactive credit account.
2	Eligibility Validation	• The system verifies whether the account meets the criteria for reinstatement. • Requests for accounts that do not meet eligibility conditions are rejected.
3	Information Submission	The applicant provides supporting documents and updates any required information necessary to justify reinstatement.
4	Validation	The system validates submitted information and documents to ensure completeness and correctness. Invalid submissions are returned for correction.
5	Review	The reviewer evaluates the reinstatement request, ensuring that all requirements are satisfied and supporting information is sufficient.
6	Routing	The system routes the request to the appropriate approver based on configured approval workflows.
7	Outcome	• The approver determines the outcome:

Step	Workflow Lifecycle	Functional Behavior
		○ **Approve** – Account is reactivated ○ **Reject** – Request is denied and workflow terminates ○ **Return** – Request is sent back for correction or additional information • The system records all relevant workflow actions for traceability.

4.3.4. Decision Points and Validation Logic

This subsection defines the key decision points that affect progression of the credit reinstatement workflow.

Table 11 Credit Reinstatement Workflow Decision Points and Validation Logic

Decision Point	Definition
Reinstatement Eligibility	Determines whether the account qualifies for reinstatement: • If eligible, the reinstatement requests proceeds to review. • If not eligible, the reinstatement request is returned for correction.
Submission Validity	Checks completeness and correctness of submitted data: • If valid, the reinstatement requests proceeds to review. • If invalid, the reinstatement request is returned for correction.
Review Completion	Determines if the reinstatement request is sufficiently complete:

Decision Point	Definition
	• If complete, the reinstatement requests proceeds to approval routing. • If incomplete, the reinstatement request is returned for correction.
Approval Outcome	Decision by approver: • If approved, account is reactivated and request moves to **Approved** state. • If reject, the request is terminated and request moves to **Rejected** state. • If returned, the request is sent back for correction and moves to **Returned** state.

Validation Logic Summary

The following validation rules are applied before allowing the request to progress to the next workflow stage:

- Mandatory field completion for reinstatement request details
- Required document submission to justify reinstatement
- Supported file format and size checks for uploaded documents
- Eligibility validation to confirm account is in a reinstatable state
- Verification of required conditions for account reactivation
- Compliance with defined business rules for reinstatement

4.3.5. State Transitions

This subsection describes how the Credit Reinstatement Workflow transitions through system states.

Note:

Returned applications re-enter the workflow through resubmission and are processed again through validation, review, and approval as applicable.

Table 12 Credit Reinstatement Workflow State Transitions

Workflow Cycle	Current State	Trigger / Condition	Next State
Initiation	**Draft**	User initiates request	**Draft**
Validation	**Draft**	Eligibility fails	**Rejected**
Information Submission	**Draft**	User submits information	**Submitted**
Validation	**Submitted**	Validation successful	**Under Review**
Validation	**Submitted**	Validation failed	**Returned**
Review	**Under Review**	Review completed	**Pending Approval**
Review	**Under Review**	Reviewer returns	**Returned**
Routing	**Pending Approval**	Approver approves	**Approved**
Routing	**Pending Approval**	Approver rejects	**Rejected**
Routing	**Pending Approval**	Approver returns	**Returned**
Outcome	**Returned**	User resubmits	**Submitted**

4.3.6. Exception Scenarios

This subsection identifies common exception scenarios that may occur in the Credit Reinstatement Workflow and describes the expected functional response:

- **Ineligible Reinstatement** – If the account does not meet reinstatement criteria, the system rejects the request and terminates the workflow.
- **Invalid Submission** – If required information or documents are incomplete or incorrect, the system returns the request for correction.

- **Reviewer Return** – If the reviewer identifies insufficient or unclear information, the request is returned for updates.
- **Approver Return** – If further clarification is required, the request is returned to the applicant.
- **Rejected Request** – If rejected, the account remains inactive and no further processing occurs.

5. Business Rules Implementation

This section defines how business rules are applied in the functional workflows of the **CardFlow Credit Origination Platform**.

5.1. Validation Rules

This subsection describes how validation rules are applied to ensure data completeness, correctness, and readiness for processing in system workflows.

Functional Application Rules

Validation rules are enforced during the **Information Submission** and **Validation** stages of all workflows.

Table 13 Validation Rules

No.	Business Rules Implementation
BRI-VAL-01	The system shall validate that all mandatory fields are completed prior to submission.
BRI-VAL-02	The system shall validate the format and structure of input data based on predefined rules.
BRI-VAL-03	The system shall validate that all required supporting documents are uploaded.
BRI-VAL-04	The system shall validate document format and file size constraints.
BRI-VAL-05	The system shall prevent progression of requests that fail validation checks.
BRI-VAL-06	The system shall return invalid requests to the initiating user with appropriate validation messages.

Workflow Impact

- **Failed Validation** – Request transitions to **Returned** state.
- **Successful Validation** – Request proceeds to **Review** stage.

5.2. Eligibility Rules

This subsection defines how eligibility rules are applied to determine whether a request can proceed in the workflow.

Eligibility conditions define whether a request or account is allowed to proceed within the workflow based on system state, user permissions, and applicable business rules:

- **Amendment Eligibility** – An account is considered eligible for amendment if:
  - It is in a modifiable state
  - It is not currently undergoing an approval process
  - The requesting user has the necessary permissions to perform updates in accordance with defined business rules
- **Reinstatement Eligibility** – An account is considered eligible for reinstatement if:
  - It is in an inactive or suspended state
  - Not permanently closed
  - Satisfies all conditions required for reactivation as defined by business rules.

Functional Application Rules

Eligibility rules are applied during the Initiation and early **Validation** stages of amendment and reinstatement workflows.

Table 14 Eligibility Rules

No.	Business Rules Implementation
BRI-ELG-01	The system shall verify eligibility of applications, amendment requests, and reinstatement requests based on defined business conditions.
BRI-ELG-02	The system shall restrict amendment requests to records that are ineligible states for modification.
BRI-ELG-03	The system shall restrict reinstatement requests to accounts that are in suspended or inactive status.
BRI-ELG-04	The system shall reject requests that do not meet eligibility criteria.

Workflow Impact

- **Failed Eligibility** – Request transitions to **Rejected** state
- **Successful Eligibility** – Request proceeds to Information Submission stage

5.3. Approval Rules

This subsection defines how approval decisions are governed in the system, including decision authority and workflow outcomes.

Functional Application Rules

Approval rules are applied during the **Review** and **Routing** stages of all workflows.

Table 15 Approval Rules

No.	Business Rules Implementation
BRI-APR-01	The system shall route requests through predefined approval workflows based on configured rules.
BRI-APR-02	The system shall determine approvers based on roles, approval thresholds, or organizational hierarchy.
BRI-APR-03	The system shall allow approvers to approve, reject, or return requests.
BRI-APR-04	The system shall require approvers to provide remarks when rejecting or returning requests.
BRI-APR-05	The system shall prevent further processing of rejected requests.

Workflow Impact

- **Approved** – Request transitions to **Approved** state
- **Rejected** – Request transitions to **Rejected** state
- **Returned** – Request transitions to **Returned** state

5.4. Routing Logic

This subsection describes how the system determines the routing of requests between reviewers and approvers based on configured workflow definitions.

Functional Application Rules

Routing logic is applied during the **Review** and **Routing** stages of workflows.

Table 16 Routing Logic

No.	Business Rules Implementation
BRI-RTG-01	The system shall route requests to reviewers and approvers based on configured workflow definitions.
BRI-RTG-02	The system shall support multi-level approval workflows.
BRI-RTG-03	The system shall determine routing paths based on predefined criteria such as roles, thresholds, or organizational hierarchy.
BRI-RTG-04	The system shall allow dynamic assignment of tasks to appropriate users.

Workflow Impact

- Determines progression from **Review** to **Pending Approval**
- Ensures correct assignment of reviewers and approvers
- Supports multi-level processing

6. Data Handling

This section describes how data is captured, processed, validated, and utilized in the workflows of the **CardFlow Credit Origination Platform**.

6.1. Data Inputs and Outputs

This subsection defines the data inputs required for processing and the outputs generated at each stage of the workflow lifecycle.

6.1.1. Data Inputs

The system captures the following types of input data:

- **Applicant Information** – Personal or organizational details required for credit evaluation
- **Application Details** – Financial, business, and supporting information relevant to the request
- **Supporting Documents** – Uploaded files used for validation and assessment
- **User Actions** – Inputs provided by reviewers and approvers during processing

6.1.2. Data Outputs

The system produces the following outputs during workflow processing:

- **Application Status Updates** – Reflects progression through workflow stages
- **Validation Results** – Indicates whether submitted data meets requirements
- **Review and Approval Decisions** – Outcomes recorded by reviewers and approvers
- **Notifications** – System-generated messages informing users of workflow events
- **Audit Records** – Logs of user actions and system events

6.1.3. Workflow Alignment

- Data inputs are captured during **Information Submission**
- Data outputs are generated across **Validation**, **Review**, **Routing**, and **Outcome** stages

6.2. Data Validation Flow

This subsection describes how data validation is applied in workflows to ensure that submitted information meets system requirements before progressing to subsequent stages.

6.2.1. Validation Flow

The following flow outlines the validation flow applied to submitted data:

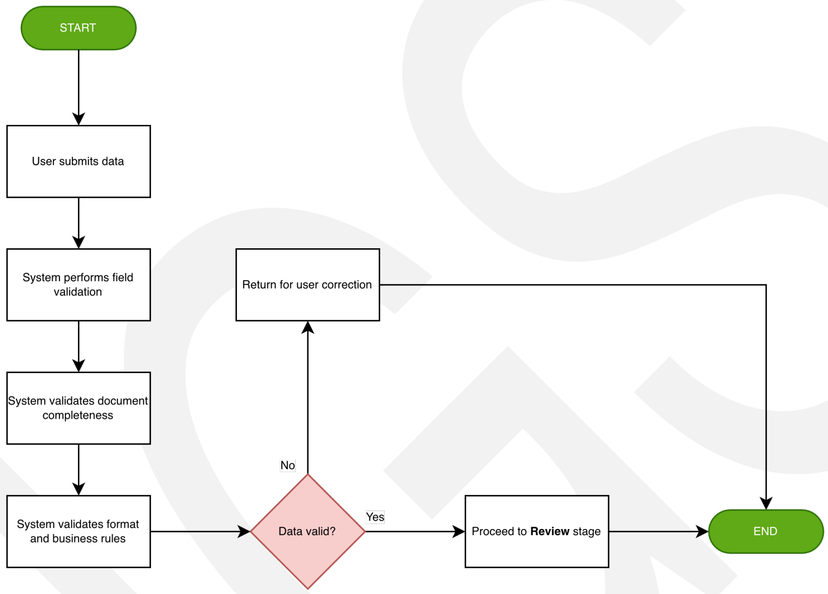

Figure 5 Data Validation Flow

6.2.2. Validation Behavior

The system performs the following validation checks on submitted data and documents.

- Mandatory fields are checked for completeness.
- Data formats are validated against predefined rules.
- Documents are verified for required types, formats, and size constraints.
- Business rule validation is applied before allowing progression.

6.2.3. Workflow Impact

The following outlines how validation results affect workflow progression:

- **Invalid Data** – Request is returned to **Information Submission**

- **Valid Data** – Request progresses to **Review**

6.3. Data Transformation In Workflows

This subsection describes how data is processed and transformed as it moves through different stages of the workflow lifecycle.

6.3.1. Transformation Behavior

As requests progress through the workflow, the system performs the following transformations:

- Submitted data is structured into application records.
- Supporting documents are linked to corresponding requests.
- Validation results are recorded and associated with the request.
- Review comments and decisions are appended to the application record.
- Approval outcomes update the final state of the request.

6.3.2. Data Enrichment

The system enriches data during processing:

- Adds workflow status and timestamps
- Records user actions and decision history
- Associates routing and approval information

6.3.3. Workflow Alignment

The following describes how data evolves as it progresses through each stage of the workflow lifecycle:

- Data is initially captured during **Information Submission**
- Data is validated during **Validation**
- Data is enriched during **Review** and **Routing**
- Final state is recorded during **Outcome**

7. User Interaction and Functional Behavior

This section describes how users interact with the system and how the system responds to user actions in the **CardFlow Credit Origination Platform**.

7.1. User Actions per Role

This subsection defines the key actions performed by each user role in the system workflows.

Table 17 User Actions per Role

Roles	Actions
Applicant	• Initiates credit application, amendment, or reinstatement requests • Provides required information and uploads supporting documents • Submits requests for processing • Updates and resubmits returned requests
Tenant User	• Initiates requests on behalf of applicants • Facilitates application or request creation
Reviewer	• Accesses submitted requests • Reviews application details and supporting documents • Determines completeness of information • Returns requests for correction when necessary
Approver	• Evaluates requests routed for approval • Approves, rejects, or returns requests

Roles	Actions
	• Provides decision remarks
System (CCOP)	• Validates submitted data and documents • Routes requests based on workflow configuration • Updates request status • Generates notifications and audit records

7.2. System Responses

This subsection describes how the system responds to user actions and workflow events.

7.2.1. System Behavior

The system responds to user actions as follows:

- **Submission of Requests** – System validates data and initiates workflow processing.
- **Validation Failure** – System returns request with validation messages.
- **Successful Validation** – System routes request to reviewer.
- **Review Completion** – System routes request to approver.
- **Approval Decision** – System updates request status and records outcome.
- **Returned Request** – System enables user to update and resubmit.

7.2.2. Response Characteristics

The following characteristics describe how the system delivers responses to user actions and workflow events:

- Responses are immediate and reflect the current workflow state.
- Responses include status updates and relevant feedback.
- Responses are logged for audit and traceability.

7.3. Role-Based Functional Access

This subsection defines how system functionalities are made available to users based on their assigned roles.

7.3.1. Access Behavior

The system enforces role-based access to ensure that users can only perform actions relevant to their responsibilities.

7.3.2. Access Rules

The following rules define how system access is controlled based on user roles:

- Applicants can create, view, and update their own requests.
- Tenant Users can initiate and manage requests in their scope.
- Reviewers can access requests assigned for review.
- Approvers can access requests pending approval.
- Administrative users can configure system settings and workflows.

7.3.3. Functional Impact

The following points describe the impact of role-based access on system functionality and workflow integrity:

- Ensures controlled access to system functionalities
- Prevents unauthorized actions
- Supports workflow integrity and security

8. Error Handling and Exception Flows

This section defines how errors and exceptional conditions are handled in the workflows of the **CardFlow Credit Origination Platform**.

8.1. Validation Failures

This subsection describes how the system handles errors related to invalid or incomplete user inputs during data submission and validation stages.

Error Handling Behavior

- The system detects incomplete or invalid data during submission.
- The system prevents progression of requests that fail validation checks.
- The system displays validation messages indicating required corrections.
- The system returns the request to the initiating user

Workflow Impact

- Request transitions to **Returned** state.
- User updates and resubmits the request.

8.2. Workflow Exceptions

This subsection describes how the system handles exceptions that occur during workflow processing, including **Review** and **Approval** stages.

Exception Handling Behavior

- The system allows reviewers to return requests when additional information is required.
- The system allows approvers to return requests for clarification or correction.
- The system ensures that returned requests can be updated and resubmitted.
- The system maintains workflow continuity without data loss.

Workflow Impact

- Request transitions to **Returned** state.
- Request re-enters workflow upon resubmission.

8.3. System Errors

This subsection describes how the system handles unexpected system-level errors during processing.

Error Handling Behavior

- The system detects unexpected errors during processing.
- The system displays user-friendly error messages without exposing sensitive details.
- The system logs errors for monitoring and investigation.
- The system ensures that incomplete operations do not result in data inconsistency.

Functional Impact

- Prevents corruption of application data.
- Supports recovery and troubleshooting.
- Maintains system reliability.

8.4. Retry and Recovery Mechanisms

This subsection describes how the system supports recovery from errors and allows retry of failed operations.

Recovery Behavior

- The system allows users to retry failed operations where applicable.
- The system enables resubmission of returned requests after correction.
- The system ensures that retried operations do not create duplicate or inconsistent data.

Functional Impact

- Ensures continuity of workflow processing
- Minimizes disruption to users
- Maintains data integrity

9. Audit and Logging Behavior

This section describes how audit and logging mechanisms are applied in the **CardFlow Credit Origination Platform** to ensure traceability of user actions and system events.

9.1. Audit Triggers

This subsection defines the events that trigger the creation of audit records in system workflows.

Audit Trigger Events

Audit records are generated when the following events occur:

- Creation of credit application, amendment, or reinstatement requests
- Submission of requests for processing
- Validation outcomes, including success and failure
- Review actions performed by reviewers
- Approval decisions made by approvers
- Return of requests for correction
- Updates and resubmission of returned requests
- Administrative configuration changes

Functional Impact

- Ensures traceability of all workflow activities
- Provides visibility into user actions and system processing
- Supports audit and compliance requirements

9.2. Logged Events per Workflow

This subsection defines the information captured in audit logs for each workflow event.

Logged Event Details

For each workflow event, the system records:

- **User Identity** – Who performed the action
- **Action Performed** – Type of activity executed
- **Timestamp** – When the action occurred

- **Affected Record** – Application or request involved
- **Previous and Updated Status** – State transition information

Workflow Alignment

Audit logging is applied across all stages:

- **Initiation** – Request creation
- **Information Submission** – Data entry and submission
- **Validation** – Validation outcomes
- **Review** – Reviewer actions
- **Routing** – Assignment of tasks
- **Outcome** – Final decisions

9.3. Traceability Points

This subsection describes how audit logs support traceability and accountability across workflows.

Traceability Behavior

- Each request maintains a complete history of actions and status changes
- All workflow transitions are recorded and linked to user actions
- Returned and resubmitted requests retain historical context
- Approval decisions are traceable to specific users and timestamps

Functional Impact

- Enables end-to-end tracking of requests
- Supports investigation and issue resolution
- Ensures accountability for decisions and actions
- Provides data for reporting and compliance

10. Integration Behavior

This section describes how the **CardFlow Credit Origination Platform** interacts with external systems and services to support data exchange, validation, and workflow processing.

10.1. External System Interaction Points

This subsection defines the points in system workflows where interaction with external systems may occur.

Interaction Points

The system may interact with external systems at the following stages:

- **Validation Stage** – Retrieval or verification of customer-related data
- **Review Stage** – Access to external reference or supporting information
- **Approval Stage** – Integration with external approval or authorization systems
- **Notification Stage** – Delivery of messages through external communication services

Functional Impact

- Enhances data accuracy through external validation
- Supports integration with enterprise systems
- Enables extended workflow capabilities beyond the platform

10.2. API Trigger Points

This subsection defines how and when APIs are invoked during workflow processing.

API Trigger Behavior

APIs may be triggered by the system during the following events:

- **Submission of Requests** – To transmit application data
- **Validation Processing** – To verify external data sources
- **Status Updates** – To synchronize workflow states with external systems
- **Approval Decisions** – To notify or update external systems

Functional Impact

- Enables real-time data exchange
- Supports system interoperability

- Ensures synchronization between internal and external systems

10.3. Integration Failure Handling

This subsection describes how the system handles failures in external system interactions.

Failure Handling Behavior

- The system detects failures in communication with external systems.
- The system logs integration errors for monitoring and troubleshooting.
- The system prevents invalid or incomplete external data from affecting workflow processing.
- The system allows retry of failed integration calls where applicable.

Functional Impact

- Maintains workflow continuity despite integration issues
- Prevents data inconsistency
- Supports recovery and error resolution